

Executive A/B Test Report

Bayesian Experiment Analysis & Verdict

2026-07-13

Executive Verdict

Result: Variation B Wins

Bayesian Probability: 100% Chance Variation B is Better.

CRITICAL RISK ALERTS

Guardrail Failure: Variation B won the primary test, but caused a statistically significant spike in negative events. Deployment is highly risky.

Simpson's Paradox: Variation B won overall, but Variation A won on 14 out of 14 days. The overall mathematical verdict is highly misleading due to traffic allocation skew.

1. Experiment Overview

Metric	Variation A (Control)	Variation B (Challenger)
Total Visitors	111,757	182,721
Total Conversions	18,671	38,006
Observed Rate	16.71%	20.8%
95% Credible Interval	[16.49% - 16.93%]	[20.61% - 20.99%]

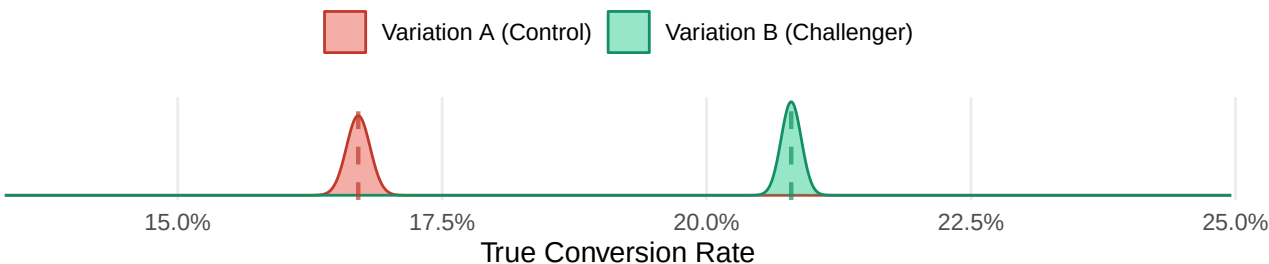
Relative Uplift: + 24.5%

Risk of Choosing B: 0%

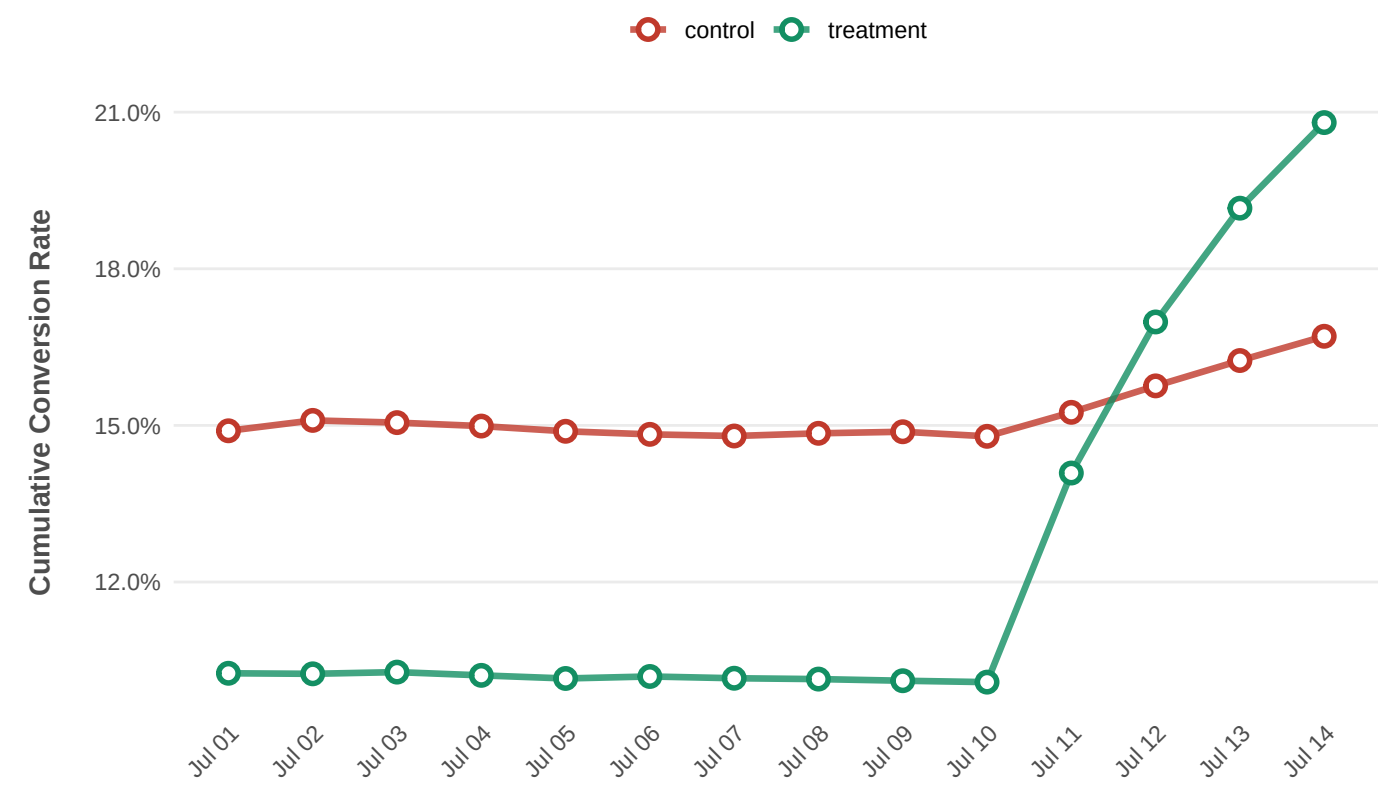
2. Posterior Probability Distributions

Insight Summary: Based on the current sample of 294,478 total visitors, Variation B is observing a conversion rate of 20.8%. At this stage, the Bayesian engine calculates a 100% probability that Variation B is the true winner. This result is statistically significant. We recommend deploying Variation B to 100% of live traffic.

The chart below visualizes the expected true conversion rates. The peak of each curve represents the most likely conversion rate for that variation based on the observed data.



3. Daily Conversion Trends



4. Power Analysis & Timeline Verdict

Test is Fully Powered!

You have enough data to make a confident decision.

Data Check: You have 294,478 total visitors. You only needed 63,852 to detect a 5 % change.

Next Step: Check the Executive Dashboard. If the result is inconclusive, it means the variation truly failed to move the needle. Shut the test down.

Appendix: Statistical Methodology

This report utilizes a Bayesian statistical framework to evaluate conversion uplift. Posterior probabilities are derived using Beta distributions modeled from an empirical prior (weighted to the pooled baseline conversion rate). The probability of Variation B outperforming Variation A is calculated via 100,000-iteration Monte Carlo simulations. The 95% credible intervals represent the mathematical bounds of the true conversion rate. Power analysis timelines are calculated assuming a two-sided test with 80% statistical power and a 5% significance level.